

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf\_2a64bc88-2a8\agent-a9c7f5725a9869c28.jsonl`
- SHA-256: `41b25219273bcd2b98fc9a1aad4074fb32ec96b4f218b54bdcaef5c336fb89b0`
- Source modified: `2026-06-10T05:41:08+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf\_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T05:33:14.934Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

TASK: deep TESTING audit of the INDEXER repo. Steps:

1. Inventory: Glob tests/** and map every test file to the backend module(s) it covers. Read pytest.ini, run_all_tests.py, TEST_RUN_SUMMARY.md, and any .github/workflows/ (does CI exist at all? check with Glob).
 2. RUN the suite: from C:/Users/avidu/Projects/badminton-highlight-indexer run "python run_all_tests.py" (offline/mocked, safe). Capture the real pass/fail/skip counts and runtime. If it fails, report exactly what failed.
 3. Coverage-gap analysis: which backend modules have NO test file at all (Glob backend/**/*.py vs tests)? Which critical paths are untested ? FastAPI endpoints (/api/process, /api/compile, /api/query), stitcher/compiler ffmpeg interaction, WSL runners (wasb_runner/tracknet_runner), database migration ladder, config loader edge cases, reporting/harvest+spec.yaml rule firing?
 4. Test QUALITY: sample ~10 test files across subsystems and judge assertion depth (smoke vs behavioral), mocking strategy (are mocks faithful to real contracts? e.g. provider analyze_video tuple shape, ffmpeg output), fixture realism, the faq-resolution golden test, regression-gate tests.
 5. Test infra: marker/selection hygiene, runtime, flakiness risk (tmp paths, ordering, global registry mutation between tests), whether tests would catch the known copy-paste drift between wasb_hybrid/tracknet_hybrid.
- Report real numbers (file counts, test counts, runtime). Severity-rate gaps by how likely they let a real regression ship.

2. user (2026-06-10T05:33:21.921Z)

No files found

3. user (2026-06-10T05:33:22.850Z)

No files found

4. user (2026-06-10T05:33:23.330Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\pytest.ini

5. user (2026-06-10T05:33:23.800Z)

No files found

6. assistant (2026-06-10T05:33:28.555Z)

The initial globs came back empty except pytest.ini ? let me check the actual repo layout.

7. user (2026-06-10T05:33:30.202Z)

```
CLAUDE.md
README.md
TEST_RUN_SUMMARY.md
__pycache__
backend
config.json
docs
output
pytest.ini
requirements.txt
run_all_tests.py
run_eval.py
run_phase3_eval.py
run_process_and_stitch.py
run_regression.py
scratch
scripts
setup.py
tests
```

8. user (2026-06-10T05:33:30.272Z)

```
1     [pytest]
2     # When `pytest` is invoked without an explicit path, only collect from tests/.
3     # scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.)
4     # and should never run as part of the regular test suite.
5     testpaths = tests
6
```

9. user (2026-06-10T05:33:37.647Z)

```
__init__.py
__pycache__
test_annotation_providers.py
test_automated_provider.py
test_base.py
test_calibrate_local.py
test_calibrate_wasb.py
test_calibration.py
test_config.py
test_database.py
test_distill_local.py
test_eval_annotations.py
test_eval_batch.py
test_eval_classifier.py
test_eval_harness.py
```

test_eval_metrics.py
test_eval_partial_labels.py
test_eval_regression.py
test_eval_training.py
test_export_wasb_segments.py
test_gemini.py
test_gemini_refine.py
test_geometry.py
test_hashing.py
test_hit_detector.py
test_provider_gemini.py
test_providers.py
test_rally_gate.py
test_rally_seq_proto.py
test_rally_slicer.py
test_reporting.py
test_reporting_harvest.py
test_reporting_spec_faq.py
test_sport_profiles.py
test_tracknet_hybrid.py
test_tracknet_runner.py
test_validators_base.py
test_video.py
test_wasb_cache.py
test_yolo_hybrid.py

40

10. user (2026-06-10T05:33:38.348Z)

/c/Users/avidu/Projects/badminton-highlight-indexer/backend/annotations/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/annotations/automated_provider.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/annotations/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/annotations/file_provider.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/config/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/config/loader.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/config/models.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/annotations.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/audio/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/audio/e1_probe.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/batch.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_local.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_wasb.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibration.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/classifier.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/distill_local.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/eval_partial_labels.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/gemini_refine.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/harness.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/metrics.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/rally_seq_proto.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/regression.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/training.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/infrastructure/database.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/main.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/audio/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/audio/hit_detector.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/rally_gate.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/tracknet_runner.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/wasb_infer.py

```

/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/detectors/wasb_runner.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/gemini.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/motion.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/tracknet_hybrid.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/wasb_hybrid.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/segmenters/yolo_hybrid.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/stitcher/compiler.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/blur.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/format.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/pts_continuity.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/relevance.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/resolution_fps.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/pipeline/validators/scene_cut.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/providers/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/providers/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/providers/gemini.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/reporting/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/reporting/harvest.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/results.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/sports/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/sports/badminton.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/sports/base.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/tools/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/tools/export.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/tools/rally_slicer.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/utils/__init__.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/utils/geometry.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/utils/hashing.py
/c/Users/avidu/Projects/badminton-highlight-indexer/backend/utils/video.py

```

11. assistant (2026-06-10T05:33:43.723Z)

Now reading the test runner, summary doc, and checking for CI.

12. user (2026-06-10T05:33:44.206Z)

```

1      #!/usr/bin/env python
2      """Run the FULL test suite for the badminton-highlight-indexer.
3
4      A single, discoverable entry point so you (or a reviewer / CI) can verify the
5      whole suite before opening a PR:
6
7          python run_all_tests.py          # full suite, quiet
8          python run_all_tests.py -v      # verbose (passthrough to pytest)
9          python run_all_tests.py -k config # filter by keyword (passthrough)
10         python run_all_tests.py --cov=backend --cov-report=term-missing # coverage
11
12     Anything after the script name is passed straight through to pytest.
13
14     Notes
15     -----
16     - The suite is fully mocked (no GPU, no network, no WSL) and runs offline.
17     - The observability **reporting** tests skip cleanly if the optional ``obsreport``
18       package isn't installed (they use ``pytest.importorskip``). To exercise them too:
19       pip install -e ../sports-obsreport --no-build-isolation
20     - Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all green).
21     """
22     from __future__ import annotations

```

```

23
24     import importlib.util
25     import subprocess
26     import sys
27
28
29     def _installed(module: str) -> bool:
30         try:
31             return importlib.util.find_spec(module) is not None
32         except (ImportError, ValueError):
33             return False
34
35
36     def main() -> int:
37         args = list(sys.argv[1:])
38         # Default to quiet unless the caller chose a verbosity.
39         if not any(a in ("-q", "-v", "-vv", "-vvv", "--quiet", "--verbose") for a in args):
40             args = ["-q", *args]
41         cmd = [sys.executable, "-m", "pytest", *args]
42
43         obs = "available" if _installed("obsreport") else "NOT installed ? reporting tests
will SKIP"
44         bar = "=" * 72
45         print(bar)
46         print(" Full test suite: " + " ".join(cmd))
47         print(f" obsreport (optional reporting dep): {obs}")
48         print(bar, flush=True)
49
50         rc = subprocess.run(cmd).returncode
51
52         print(bar)
53         print(" RESULT: " + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}"))
54         print(bar)
55         return rc
56
57
58     if __name__ == "__main__":
59         raise SystemExit(main())
60

```

13. user (2026-06-10T05:33:45.152Z)

```

1     # Test Run Diagnosis & Optimizations Summary
2
3     > **Historical note:** This is a point-in-time diagnosis from a May 2026 local test run,
4     > written *before* the layered-directory restructure (PR #40). Path references below
5     > reflect the old flat layout (e.g. a single `segmenter.py`, now the
6     > `backend/pipeline/segmenters/` package). The diagnostic findings and optimization
7     > lessons remain valid; treat the file paths and install commands as historical.
8
9     *Date: May 18, 2026*
10    *Target File: `C:\Users\avidu\Videos\LargeTestVideo.MP4` (HEVC H.265, Size: ~11.5 GB)*
11
12    This summary details the findings, architectural lessons, and performance optimizations
13    implemented during today's local test run of the Badminton Highlight & Rally Indexer.
14
15    ---
16
17    ## ? Why the Initial Test Run Blocked (and How We Resolved It)
18
19    During the raw test execution on the 11.5 GB video stream, the system hit three
20    distinct, high-impact constraints. We fully diagnosed, optimized, and hardened the pipelines to
21    address them:
22

```

```

20     ### 1. The Keyframe Extraction Gap (H.265 Stream Header Parsing)
21     *   **The Issue**: The `PTSContinuityValidator` check initially failed with `[FAIL] No
keyframes could be extracted`.
22     *   **The Root Cause**: GoPro raw H.265/HEVC encodes use **Gradual Decoder Refresh
(GDR)** or **Clean Random Access (CRA)** keyframe configurations. Standard FFmpeg frame-level
decoding with `-skip_frame nokey` relies on hardware/software decoders which failed to flag
standard keyframe structures, returning empty outputs.
23     *   **The Optimization**: We changed the validator to query packet indexes
(`packet=pts_time,flags`) at the **container demuxer level** instead of full frame decoding.
Because packet flags (e.g. `K__`) are written directly to container headers, this bypasses the
video decoder entirely. It runs **100x faster** and is 100% robust across HEVC containers!
24
25     ### 2. The Blurriness Quality Barrier (Indoor Arena Low-Light Noise)
26     *   **The Issue**: The video failed the focus threshold check with a sharpness score of
`35.3` (minimum threshold was `100.0`).
27     *   **The Root Cause**: High-speed action footage in typical indoor badminton
environments suffers from high-speed motion blur and ISO sensor noise, reducing the computed
Laplacian variance.
28     *   **The Fix**: We lowered the `min_blur_threshold` default inside `config.json` to
`20.0` so that raw fast-moving arena footage passes quality checks cleanly.
29
30     ### 3. OpenCV Seeking Performance Bottleneck
31     *   **The Issue**: The segmenter process consumed massive CPU time during early stages,
appearing to respond slowly.
32     *   **The Root Cause**: Seeking to arbitrary frames in highly compressed, high-bitrate
large H.265/HEVC files using OpenCV's `cap.set(cv2.CAP_PROP_POS_FRAMES)` is an extremely
expensive seek operation. Because the decoder has to seek to the closest keyframe and decode
forward to the target frame for every sub-sampled frame, it created an  $O(N^2)$  frame-decoding
loop.
33     *   **The Optimization**: We refactored the segmenter (then `segmenter.py`, now the
`backend/pipeline/segmenters/` package) to use a high-performance **sequential grab-and-retrieve
pipeline**. By skipping intermediate frames using `cap.grab()` (which parses container
structures without decoding pixel values) and only calling `cap.read()` for our 5 FPS analysis
frames sequentially, we avoided seeks entirely. This delivers a **100x speedup** on multi-
gigabyte MP4s!
34
35     ---
36
37     ## ?? Newly Added Debug Options: `--max-frames`
38
39     To allow you to quickly test video subsets without waiting to process an entire multi-
gigabyte video file, we added the `--max-frames <N>` argument (and `max_frames` in the API
payload):
40     *   This restricts the segmenter frame loop to analyze exactly the first  $N$  sub-sampled
frames (at 5 FPS, `--max-frames 300` executes only the first 60 seconds of play).
41     *   It immediately exits the loop at the limit and proceeds to segment, cluster, and
seed the database with whatever play timelines it identified in that subset.
42     *   This lets you debug whether your visual indices and relevance checks are executing
correctly in **seconds** instead of minutes!
43
44     ---
45
46     ## ? Guidelines for Running on a More Powerful Machine
47
48     Checking this repository into your private GitHub will let you execute the indexer on a
high-powered machine. Here is how to maximize performance:
49
50     1. **CUDA/GPU Hardware Acceleration**:
51         If your other machine has a CUDA-enabled NVIDIA GPU, install PyTorch with CUDA:
52         ```bash
53         pip install torch torchvision --index-url https://download.pytorch.org/whl/cu124
54         ```
55         (`setup.py` auto-selects the matching CUDA wheel ? currently `cu124` ? when it
detects an NVIDIA GPU.)
56         Our diagnostics checker (`setup.py`) will automatically detect CUDA and leverage

```

```
standard GPU matrix operations to perform accelerated image difference thresholding!
57      2.  **Unbuffered Console Outputs**:
58          To see log statements instantly without OS shell output buffers, use Python's
unbuffered command:
59          ```bash
60          python -u -m backend.main --video-path "C:\Users\avidu\Videos\LargeTestVideo.MP4"
--max-frames 500
61          ```
62
```

14. user (2026-06-10T05:33:46.270Z)

```
exit=1
```

15. user (2026-06-10T05:34:19.644Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 28%]
..... [ 56%]
..... [ 85%]
..... [100%]
254 passed in 19.90s
=====
RESULT: PASS - all tests green
=====

real    0m22.719s
user    0m0.031s
sys     0m0.078s
```

16. user (2026-06-10T05:34:42.870Z)

```
test_annotation_providers.py -> backend.annotations import annotation_registry,
AnnotationProvider
test_annotation_providers.py -> backend.annotations.base import DONE, FAILED
test_annotation_providers.py -> backend.annotations.file_provider import FileProvider
test_automated_provider.py -> backend.annotations import annotation_registry
test_automated_provider.py -> backend.annotations.automated_provider import (
test_automated_provider.py -> backend.annotations.base import DONE, FAILED
test_base.py -> backend.pipeline.segmenters.base import BasePlaySegmenter, SegmenterRegistry
test_base.py -> backend.results import Success, Failure, Result # test the new contract
test_calibrate_local.py -> backend.eval import calibrate_local as cl
test_calibrate_wasb.py -> backend.eval.calibrate_wasb import (
test_calibration.py -> backend.eval.calibration import (
test_config.py -> backend.config import load_config, save_default_config, AppConfig
test_database.py -> backend.infrastructure.database import Database
test_distill_local.py -> backend.eval import distill_local as dl
test_distill_local.py -> backend.eval.classifier import LogisticRegression
test_eval_annotations.py -> backend.eval.annotations import parse_timestamp, load_annotations,
write_scaffold
test_eval_batch.py -> backend.eval import batch
test_eval_classifier.py -> backend.eval.classifier import LogisticRegression
test_eval_harness.py -> backend.eval.harness import (
test_eval_harness.py -> backend.infrastructure.database import Database
test_eval_metrics.py -> backend.eval.metrics import (
test_eval_partial_labels.py -> backend.eval import eval_partial_labels as ep
test_eval_regression.py -> backend.eval import regression
test_eval_regression.py -> backend.infrastructure.database import Database
test_eval_training.py -> backend.eval import training
test_export_wasb_segments.py -> backend.eval import export_wasb_segments as ex
```

```

test_export_wasb_segments.py -> backend.tools.rally_slicer import load_rally_labels
test_gemini.py -> backend.pipeline.segmenters.gemini import GeminiPlaySegmenter
test_gemini_refine.py -> backend.eval import gemini_refine as gr
test_geometry.py -> backend.utils.geometry import get_center, get_distance, get_velocity,
get_velocity_normalised, calculate_iou
test_hashing.py -> backend.utils.hashing import compute_video_id
test_hit_detector.py -> backend.eval.audio.e1_probe import cluster_hits
test_hit_detector.py -> backend.pipeline.audio import hit_detector as hd
test_hit_detector.py -> backend.pipeline.audio.hit_detector import HitEvent
test_provider_gemini.py -> backend.providers.gemini import GeminiProvider
test_providers.py -> backend.providers import provider_registry
test_providers.py -> backend.providers.base import AIVideoProvider
test_providers.py -> backend.providers.gemini import GeminiProvider
test_rally_gate.py -> backend.pipeline.detectors import rally_gate as rg
test_rally_gate.py -> backend.pipeline.detectors.tracknet_runner import TrajectoryPoint
test_rally_seq_proto.py -> backend.eval import rally_seq_proto as rp
test_rally_slicer.py -> backend.tools.export import basic_stats, segments_to_json,
segments_to_csv, segments_to_edl
test_rally_slicer.py -> backend.tools.rally_slicer import (
test_reporting.py -> backend.reporting import emit_indexer_report, report_enabled
test_reporting.py -> backend.reporting.harvest import video_meta_from_context # pure helper; no
obsreport dep
test_reporting_harvest.py -> backend.pipeline.validators.base import ValidationResult # noqa:
E402
test_reporting_harvest.py -> backend.reporting import emit_indexer_report, harvest # noqa: E402
test_sport_profiles.py -> backend.sports import sport_registry
test_sport_profiles.py -> backend.sports.badminton import BadmintonSportProfile
test_sport_profiles.py -> backend.sports.base import SportProfile
test_tracknet_hybrid.py -> backend.pipeline.segmenters.tracknet_hybrid import
TrackNetHybridSegmenter
test_tracknet_runner.py -> backend.pipeline.detectors.base import DetectorRunner
test_tracknet_runner.py -> backend.pipeline.detectors.tracknet_runner import (
test_tracknet_runner.py -> backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
test_validators_base.py -> backend.pipeline.validators.base import ValidationRegistry,
ValidationResult, VideoValidator
test_video.py -> backend.utils.video import get_video_duration, extract_video_chunk
test_wasb_cache.py -> backend.pipeline.detectors import wasb_infer as wi
test_yolo_hybrid.py -> backend.pipeline.segmenters.yolo_hybrid import HybridYoloSegmenter

```

17. user (2026-06-10T05:34:54.164Z)

```

tests/test_yolo_hybrid.py::test_yolo_hybrid_persists_and_links_candidates
tests/test_yolo_hybrid.py::test_yolo_hybrid_store_candidates_disabled
tests/test_yolo_hybrid.py::test_detect_and_track_filters_persons_and_returns_track_ids

```

```

254 tests collected in 7.19s
21 tests/test_tracknet_runner.py
17 tests/test_eval_metrics.py
14 tests/test_eval_harness.py
12 tests/test_rally_slicer.py
11 tests/test_wasb_cache.py
11 tests/test_reporting_harvest.py
11 tests/test_eval_annotations.py
9 tests/test_eval_regression.py
9 tests/test_database.py
9 tests/test_calibration.py
9 tests/test_automated_provider.py
9 tests/test_annotation_providers.py
8 tests/test_tracknet_hybrid.py
8 tests/test_eval_batch.py
7 tests/test_yolo_hybrid.py
7 tests/test_hit_detector.py
7 tests/test_eval_training.py
7 tests/test_config.py
6 tests/test_export_wasb_segments.py

```



```
6 tests/test_eval_classifier.py
5 tests/test_reporting.py
5 tests/test_rally_gate.py
5 tests/test_geometry.py
4 tests/test_video.py
4 tests/test_provider_gemini.py
4 tests/test_hashing.py
4 tests/test_gemini_refine.py
4 tests/test_eval_partial_labels.py
3 tests/test_validators_base.py
3 tests/test_rally_seq_proto.py
3 tests/test_calibrate_wasb.py
2 tests/test_sport_profiles.py
2 tests/test_reporting_spec_faq.py
2 tests/test_gemini.py
2 tests/test_distill_local.py
2 tests/test_calibrate_local.py
1 tests/test_providers.py
1 tests/test_base.py
```

18. user (2026-06-10T05:35:10.488Z)

```
49:@app.get("/", response_class=HTMLResponse)
50:def serve_index():
63:def load_config(config_path: str = "config.json") -> Dict[str, Any]:
87:@app.get("/api/config")
88:def get_config():
91:@app.post("/api/process")
92:def process_video(req: ProcessRequest):
172:@app.post("/api/query")
173:def query_play_segments(req: QueryRequest):
183:@app.post("/api/compile")
184:def compile_highlights(req: CompileRequest):
209:def run_cli_diagnose_clip(video_path: str, timestamp: float):
280:def main():
=== compiler ===
2:import subprocess
9:class VideoCompiler:
50:    result = subprocess.run(
138:        "ffmpeg", "-y",
151:        result = subprocess.run(cmd, capture_output=True, check=True)
158:    except subprocess.CalledProcessError as e:
163:        skipped_segments.append((idx, start, end, f"ffmpeg error:
{stderr_msg[-200:]})")
178:        f"Common causes: video file is corrupted, segments are out of range, or
ffmpeg is not installed."
194:        "ffmpeg", "-y",
203:        result = subprocess.run(concat_cmd, capture_output=True, check=True)
215:    except subprocess.CalledProcessError as e:
=== wc ===
426 backend/main.py
221 backend/pipeline/stitcher/compiler.py
225 backend/pipeline/segmenters/wasb_hybrid.py
254 backend/pipeline/segmenters/tracknet_hybrid.py
229 backend/pipeline/segmenters/motion.py
267 backend/infrastructure/database.py
1622 total
```

19. user (2026-06-10T05:35:22.617Z)

```
tests/test_config.py:92: runtime mutation that backend/main.py::main performs for --output-
dir."""
tests/test_hashing.py:4:(backend/main.py + the three run_*.py scripts) into one shared function.
Binary file tests/__pycache__/test_config.cpython-313-pytest-9.0.3.pyc matches
Binary file tests/__pycache__/test_hashing.cpython-313-pytest-9.0.3.pyc matches
```

```

testclient-exit=0
ls: cannot access 'tests/conftest.py': No such file or directory
=== registry mutation ===
tests/test_annotation_providers.py
tests/test_automated_provider.py
tests/test_base.py
tests/test_gemini.py
tests/test_providers.py
tests/test_sport_profiles.py
tests/test_tracknet_hybrid.py
tests/test_validators_base.py
tests/test_yolo_hybrid.py
tests/__pycache__/test_annotation_providers.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_automated_provider.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_base.cpython-311-pytest-9.0.3.pyc
tests/__pycache__/test_base.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_gemini.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_providers.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_sport_profiles.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_tracknet_hybrid.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_validators_base.cpython-313-pytest-9.0.3.pyc
tests/__pycache__/test_yolo_hybrid.cpython-313-pytest-9.0.3.pyc

```

20. user (2026-06-10T05:35:22.622Z)

```

1     import pytest
2     import sqlite3
3     import os
4     from backend.infrastructure.database import Database
5
6     def test_database_init(tmp_path):
7         db_path = tmp_path / "test.db"
8         db = Database(str(db_path))
9
10        conn = sqlite3.connect(str(db_path))
11        cursor = conn.cursor()
12        cursor.execute("SELECT name FROM sqlite_master WHERE type='table'")
13        tables = [r[0] for r in cursor.fetchall()]
14        assert "videos" in tables
15        assert "play_segments" in tables
16        conn.close()
17
18    def test_database_video(tmp_path):
19        db = Database(str(tmp_path / "test.db"))
20
21        vid_added = db.add_video("vid1", "vid1.mp4", "pending", [{"val": 1}])
22        assert vid_added is True
23
24        video = db.get_video("vid1")
25        assert video["id"] == "vid1"
26        assert video["filepath"] == "vid1.mp4"
27        assert video["status"] == "pending"
28        assert video["validation_results"] == [{"val": 1}]
29
30        db.clear_video_data("vid1")
31        assert db.get_video("vid1") is None
32
33    def test_database_play_segment(tmp_path):
34        db = Database(str(tmp_path / "test.db"))
35        db.add_video("vid1", "vid1.mp4", "pending", [])
36
37        sid = db.add_play_segment("vid1", 10.0, 20.0, "PlayerA", "PlayerB", {"foo": "bar"})
38        assert sid is not None
39
40        db.add_event(sid, 15.0, "smash", "PlayerA", {"speed": 300})

```

```

41
42     segments = db.query_play_segments("vid1", {"server": "PlayerA"})
43     assert len(segments) == 1
44     assert segments[0]["id"] == sid
45     assert segments[0]["start_time"] == 10.0
46     assert segments[0]["server"] == "PlayerA"
47     assert segments[0]["metadata"]["foo"] == "bar"
48
49     assert len(segments[0]["events"]) == 1
50     assert segments[0]["events"][0]["type"] == "smash"
51     assert segments[0]["events"][0]["details"]["speed"] == 300
52
53
54     # --- candidate_segments schema + behavior -----
55
56     # The schema contract: any future change to candidate_segments columns must
57     # consciously update this set, otherwise the test below fails (regression guard).
58     EXPECTED_CANDIDATE_COLUMNS = {
59         "id", "video_id", "detector", "chunk_index", "start_time", "end_time",
60         "duration", "confidence", "stats", "detection_params",
61         "confirmed_segment_id", "human_label", "notes", "created_at",
62     }
63
64
65     def test_candidate_schema_contract(tmp_path):
66         db = Database(str(tmp_path / "test.db"))
67         conn = sqlite3.connect(str(tmp_path / "test.db"))
68         cols = conn.execute("PRAGMA table_info(candidate_segments)").fetchall()
69         conn.close()
70
71         names = {c[1] for c in cols}
72         assert names == EXPECTED_CANDIDATE_COLUMNS
73
74         # NOT NULL / PK flags on the load-bearing columns (cid, name, type, notnull, dflt,
pk)
75         by_name = {c[1]: c for c in cols}
76         assert by_name["id"][5] == 1 # primary key
77         for col in ("video_id", "detector", "start_time", "end_time", "duration"):
78             assert by_name[col][3] == 1, f"{col} should be NOT NULL"
79
80
81     def test_schema_version_is_current(tmp_path):
82         db_path = tmp_path / "test.db"
83         Database(str(db_path))
84         conn = sqlite3.connect(str(db_path))
85         version = conn.execute("PRAGMA user_version").fetchone()[0]
86         conn.close()
87         assert version == 1
88
89
90     def test_init_db_idempotent_and_upgrades_old_db(tmp_path):
91         db_path = tmp_path / "legacy.db"
92         # Simulate a pre-v1 database that only has the old tables and no version.
93         conn = sqlite3.connect(str(db_path))
94         conn.execute("CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT, status
TEXT)")
95         conn.commit()
96         conn.close()
97
98         # Opening with the new Database should add candidate_segments without error...
99         Database(str(db_path))
100        # ...and running init again must be a no-op (idempotent).
101        Database(str(db_path))
102
103        conn = sqlite3.connect(str(db_path))

```

```

104     tables = [r[0] for r in conn.execute(
105         "SELECT name FROM sqlite_master WHERE type='table'").fetchall()]
106     version = conn.execute("PRAGMA user_version").fetchone()[0]
107     conn.close()
108     assert "candidate_segments" in tables
109     assert version == 1
110
111
112 def test_candidate_round_trip(tmp_path):
113     db = Database(str(tmp_path / "test.db"))
114     db.add_video("vid1", "vid1.mp4", "processed", [])
115
116     cid = db.add_candidate_segment(
117         "vid1", "yolo_hybrid", 10.0, 25.0, chunk_index=0, confidence=0.42,
118         stats={"peak_velocity": 0.8, "track_id_count": 2},
119         detection_params={"frame_skip": 3},
120     )
121     assert cid is not None
122
123     cands = db.query_candidate_segments("vid1")
124     assert len(cands) == 1
125     c = cands[0]
126     assert c["detector"] == "yolo_hybrid"
127     assert c["start_time"] == 10.0
128     assert c["duration"] == 15.0
129     assert c["confidence"] == 0.42
130     assert c["stats"]["peak_velocity"] == 0.8
131     assert c["detection_params"]["frame_skip"] == 3
132     assert c["confirmed_segment_id"] is None
133
134     # detector filter
135     assert db.query_candidate_segments("vid1", detector="motion") == []
136     assert len(db.query_candidate_segments("vid1", detector="yolo_hybrid")) == 1
137
138
139 def test_candidate_link_and_unlabeled_filter(tmp_path):
140     db = Database(str(tmp_path / "test.db"))
141     db.add_video("vid1", "vid1.mp4", "processed", [])
142
143     cid = db.add_candidate_segment("vid1", "yolo_hybrid", 1.0, 4.0)
144     sid = db.add_play_segment("vid1", 1.0, 4.0)
145
146     assert db.link_candidate_to_segment(cid, sid) is True
147     assert db.query_candidate_segments("vid1")[0]["confirmed_segment_id"] == sid
148
149     # only_unlabeled returns it while human_label is NULL
150     assert len(db.query_candidate_segments("vid1", only_unlabeled=True)) == 1
151     with db._get_connection() as conn:
152         conn.execute("UPDATE candidate_segments SET human_label = ? WHERE id = ?",
153             ("true_rally", cid))
154         conn.commit()
155     assert db.query_candidate_segments("vid1", only_unlabeled=True) == []
156
157
158 def test_candidate_cascade_on_video_delete(tmp_path):
159     db = Database(str(tmp_path / "test.db"))
160     db.add_video("vid1", "vid1.mp4", "processed", [])
161     db.add_candidate_segment("vid1", "yolo_hybrid", 1.0, 4.0)
162
163     db.clear_video_data("vid1")
164     assert db.query_candidate_segments("vid1") == []
165

```

```

1     import pytest
2     from unittest.mock import MagicMock, patch
3     from backend.providers.gemini import GeminiProvider
4
5     def test_gemini_provider_success():
6         # Mock GenAI client
7         with patch("backend.providers.gemini.genai") as mock_genai:
8             client_mock = MagicMock()
9             mock_genai.Client.return_value = client_mock
10
11             # Mock file upload
12             file_mock = MagicMock()
13             file_mock.uri = "gs://dummy"
14             file_mock.name = "dummy_file"
15             file_mock.state.name = "ACTIVE"
16
17             client_mock.files.upload.return_value = file_mock
18             client_mock.files.get.return_value = file_mock
19
20             # Mock generate_content response
21             response_mock = MagicMock()
22             response_mock.text = '["start_time": 1.0, "end_time": 2.0]']
23             client_mock.models.generate_content.return_value = response_mock
24
25             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
26             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt",
chunk_duration=10.0)
27
28             assert len(segments) == 1
29             assert segments[0]["start_time"] == 1.0
30             assert metrics["upload_time"] >= 0
31             assert metrics["process_time"] >= 0
32             assert metrics["gen_time"] >= 0
33
34             # Ensure delete is called
35             client_mock.files.delete.assert_called_with(name="dummy_file")
36
37     def test_gemini_provider_invalid_json():
38         with patch("backend.providers.gemini.genai") as mock_genai:
39             client_mock = MagicMock()
40             mock_genai.Client.return_value = client_mock
41             file_mock = MagicMock()
42             file_mock.state.name = "ACTIVE"
43             client_mock.files.upload.return_value = file_mock
44             client_mock.files.get.return_value = file_mock
45
46             response_mock = MagicMock()
47             response_mock.text = '{"not_a_list": true}'
48             client_mock.models.generate_content.return_value = response_mock
49
50             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
51             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")
52             assert len(segments) == 0
53
54     def test_gemini_provider_rejects_oversize_before_upload(monkeypatch):
55         with patch("backend.providers.gemini.genai") as mock_genai:
56             client_mock = MagicMock()
57             mock_genai.Client.return_value = client_mock
58             # 600MB > MAX_UPLOAD_MB (500) -> refuse without ever uploading
59             monkeypatch.setattr("backend.providers.gemini.os.path.getsize",
lambda p: 600 * 1024 * 1024)
60
61             provider = GeminiProvider(api_key="k", model_name="gemini-flash-latest")
62             segments, metrics = provider.analyze_video("big.mp4", "p")

```

```

63         assert segments == []
64         client_mock.files.upload.assert_not_called()
65
66
67     def test_gemini_provider_upload_fail():
68         with patch("backend.providers.gemini.genai") as mock_genai:
69             client_mock = MagicMock()
70             mock_genai.Client.return_value = client_mock
71             client_mock.files.upload.side_effect = Exception("Upload err")
72
73             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
74             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")
75             assert len(segments) == 0
76

```

22. user (2026-06-10T05:35:39.718Z)

```

1     import pytest
2     from unittest.mock import patch, MagicMock
3     from backend.utils.video import get_video_duration, extract_video_chunk
4
5     @patch("subprocess.check_output")
6     def test_get_video_duration_success(mock_check_output):
7         mock_check_output.return_value = b"123.45\n"
8         dur = get_video_duration("dummy.mp4")
9         assert dur == 123.45
10
11    @patch("subprocess.check_output")
12    def test_get_video_duration_failure(mock_check_output):
13        mock_check_output.side_effect = Exception("ffprobe error")
14        dur = get_video_duration("dummy.mp4")
15        assert dur == 0.0
16
17    @patch("subprocess.run")
18    def test_extract_video_chunk_success(mock_run):
19        success = extract_video_chunk("in.mp4", 10.0, 5.0, "out.mp4")
20        assert success is True
21        mock_run.assert_called_once()
22
23    @patch("subprocess.run")
24    def test_extract_video_chunk_failure(mock_run):
25        mock_run.side_effect = Exception("ffmpeg error")
26        success = extract_video_chunk("in.mp4", 10.0, 5.0, "out.mp4")
27        assert success is False
28

```

23. user (2026-06-10T05:35:48.537Z)

```

1     """Golden guard: every footgun rule's faq_ref must resolve to a real README heading.
2
3     This is the committed backbone of report debuggability ? if a rule cites README#Qn,
4     that section must exist, so a reader who follows the pointer always lands somewhere.
5     (A fresh-reader eval found a rule citing the wrong-but-existing section; an existence
6     test can't catch *semantic* mismatch, but it locks the floor: no dangling anchors.)
7     """
8     import re
9     from pathlib import Path
10
11    import pytest
12
13    pytest.importorskip("obsreport")
14    from obsreport import load_spec # noqa: E402
15
16    ROOT = Path(__file__).resolve().parent.parent

```

```

17     SPEC = ROOT / "backend" / "reporting" / "spec.yaml"
18     README = ROOT / "README.md"
19
20
21     def _headings(md: str):
22         return [ln.lstrip("#").strip() for ln in md.splitlines() if
ln.lstrip().startswith("#")]
23
24
25     def _slug(h: str) -> str:
26         s = re.sub(r"^[^w\s-]", "", h.lower())
27         return re.sub(r"\s+", "-", s.strip())
28
29
30     def _anchor_resolves(anchor: str, headings) -> bool:
31         a = anchor.lower()
32         for h in headings:
33             hl = h.lower()
34             if hl == a or hl.startswith(a + ":") or hl.startswith(a + " ") or
hl.startswith(a + "."):
35                 return True # Q-number style: "Q7" -> "### Q7: ..."
36             if _slug(h) == a:
37                 return True # full GitHub slug match
38         return False
39
40
41     def test_every_rule_faq_ref_resolves_to_a_readme_heading():
42         spec = load_spec(SPEC)
43         headings = _headings(README.read_text(encoding="utf-8"))
44         unresolved = []
45         for rule in spec.rules:
46             if not rule.faq_ref:
47                 continue
48             file_part, _, anchor = rule.faq_ref.partition("#")
49             if file_part in ("README", "README.md"):
50                 if not _anchor_resolves(anchor, headings):
51                     unresolved.append((rule.code, rule.faq_ref))
52             elif file_part in ("VIDEO_RECORDING_GUIDELINES",
"VIDEO_RECORDING_GUIDELINES.md",
53 "AUDIO_RALLY_DETECTION_PLAN",
"AUDIO_RALLY_DETECTION_PLAN.md"):
54                 # These are dedicated guidance docs surfaced from report rules (Input
Quality
55                 # extension). The whole document is the destination; no #Q anchor required.
56                 pass
57             else:
58                 unresolved.append((rule.code, rule.faq_ref))
59         assert not unresolved, f"faq_ref anchors with no matching README heading:
{unresolved}"
60
61
62     def test_actionable_rules_have_faq_refs():
63         """error/warn findings must cite a FAQ (info may omit). Belt-and-suspenders on top
64         of obsreport's own spec-load validation."""
65         spec = load_spec(SPEC)
66         missing = [r.code for r in spec.rules if r.severity in ("error", "warn") and not
r.faq_ref]
67         assert not missing, f"actionable rules missing faq_ref: {missing}"
68

```

24. user (2026-06-10T05:35:48.989Z)

```

1     """Tests for the regression runner (GS-2): baseline store, regress(), the
2     provider-driven end-to-end path, and the CLI exit-code contract."""
3     import json

```

```

4     import pytest
5
6     from backend.infrastructure.database import Database
7     from backend.eval import regression
8     import run_regression
9
10
11     @pytest.fixture
12     def db(tmp_path):
13         d = Database(str(tmp_path / "t.db"))
14         d.add_video("vid1", "/x/m.mp4", "processed", [])
15         # Two confirmed segments matching the GT below ? perfect baseline.
16         d.add_play_segment("vid1", 0.0, 10.0)
17         d.add_play_segment("vid1", 20.0, 30.0)
18         return d
19
20
21     GT = [(0.0, 10.0), (20.0, 30.0)]
22
23
24     # ----- baseline store -----
25
26     def test_save_and_load_baseline_roundtrip(tmp_path):
27         p = str(tmp_path / "b.json")
28         regression.save_baseline("vid1", "final", 1.0, 1.0, 1.0, path=p)
29         b = regression.load_baselines(p)
30         assert b["vid1::final"]["precision"] == 1.0
31
32
33     def test_load_baselines_absent_is_empty(tmp_path):
34         assert regression.load_baselines(str(tmp_path / "nope.json")) == {}
35
36
37     # ----- regress() -----
38
39     def test_no_baseline_is_not_a_regression(db, tmp_path):
40         r = regression.regress(db, "vid1", GT, baseline_path=str(tmp_path / "b.json"))
41         assert r.regressed is False
42         assert r.precision == 1.0 and r.recall == 1.0
43         assert any("no baseline" in why for why in r.reasons)
44
45
46     def test_matching_baseline_passes(db, tmp_path):
47         p = str(tmp_path / "b.json")
48         regression.save_baseline("vid1", "final", 1.0, 1.0, 1.0, path=p)
49         r = regression.regress(db, "vid1", GT, baseline_path=p)
50         assert r.regressed is False
51
52
53     def test_recall_drop_beyond_tolerance_regresses(tmp_path):
54         # DB has only ONE of the two GT rallies ? recall 0.5 vs baseline 1.0.
55         d = Database(str(tmp_path / "t.db"))
56         d.add_video("vid1", "/x/m.mp4", "processed", [])
57         d.add_play_segment("vid1", 0.0, 10.0)
58         p = str(tmp_path / "b.json")
59         regression.save_baseline("vid1", "final", 1.0, 1.0, 1.0, path=p)
60
61         r = regression.regress(d, "vid1", GT, tolerance=0.05, baseline_path=p)
62         assert r.regressed is True
63         assert r.recall == pytest.approx(0.5)
64         assert any("recall" in why for why in r.reasons)
65
66
67     def test_within_tolerance_does_not_regress(tmp_path):
68         d = Database(str(tmp_path / "t.db"))

```



```

69     d.add_video("vid1", "/x/m.mp4", "processed", [])
70     d.add_play_segment("vid1", 0.0, 10.0)
71     d.add_play_segment("vid1", 20.0, 30.0)
72     p = str(tmp_path / "b.json")
73     # Baseline only 0.02 above current perfect score ? within default 0.05 tolerance.
74     regression.save_baseline("vid1", "final", 1.0, 1.0, 1.0, path=p)
75     r = regression.regress(d, "vid1", GT, tolerance=0.05, baseline_path=p)
76     assert r.regressed is False
77
78
79     # ----- provider-driven end-to-end (uses the GS-1 FileProvider) -----
80
81     def test_regress_with_provider_file_tier(db, tmp_path):
82         csv = tmp_path / "gt.csv"
83         csv.write_text("start,end\n0,10\n20,30\n")
84         r = regression.regress_with_provider(
85             db, video_id="vid1", video_ref=str(csv), tier="file",
86             baseline_path=str(tmp_path / "b.json"),
87         )
88         assert r.precision == 1.0 and r.recall == 1.0
89
90
91     # ----- CLI exit-code contract (CI gating) -----
92
93     def test_cli_update_baseline_then_clean_run(db, tmp_path, capsys, monkeypatch):
94         # Point the runner at our temp DB via --db; CSV ground truth via 'file' tier.
95         csv = tmp_path / "gt.csv"
96         csv.write_text("start,end\n0,10\n20,30\n")
97         base = str(tmp_path / "b.json")
98         db_path = db.db_path if hasattr(db, "db_path") else str(tmp_path / "t.db")
99
100         # First, snapshot the baseline ? exit 0.
101         rc = run_regression.main(["--video-id", "vid1", "--tier", "file",
102                                 "--annotations", str(csv), "--db", db_path,
103                                 "--baseline", base, "--update-baseline"])
104         assert rc == 0
105         assert "REGRESSED" not in capsys.readouterr().out
106
107
108     def test_cli_regression_exits_nonzero(tmp_path, capsys):
109         # DB missing one rally vs a baseline of perfect ? regression ? exit 1.
110         d = Database(str(tmp_path / "t.db"))
111         d.add_video("vid1", "/x/m.mp4", "processed", [])
112         d.add_play_segment("vid1", 0.0, 10.0)
113         csv = tmp_path / "gt.csv"
114         csv.write_text("start,end\n0,10\n20,30\n")
115         base = str(tmp_path / "b.json")
116         regression.save_baseline("vid1", "final", 1.0, 1.0, 1.0, path=base)
117
118         rc = run_regression.main(["--video-id", "vid1", "--tier", "file",
119                                 "--annotations", str(csv), "--db", str(tmp_path / "t.db"),
120                                 "--baseline", base])
121         assert rc == 1
122         assert "REGRESSED" in capsys.readouterr().out
123

```

25. user (2026-06-10T05:35:56.321Z)

```

1     """Tests for the TrackNetHybridSegmenter pipeline. The WSL/TensorFlow runner is
2     mocked out (TrackNetRunner), so these run anywhere ? they exercise Phase 1-4 glue:
3     healthcheck gating, candidate persistence, AI handoff, and DB population/linking."""
4
5     import cv2
6     import pytest
7     from unittest.mock import MagicMock, patch

```

```

8
9     from backend.pipeline.segmenters.tracknet_hybrid import TrackNetHybridSegmenter
10
11
12     def _window(start=1.0, end=4.0):
13         return {
14             "start": start, "end": end, "active_frame_count": 30,
15             "peak_velocity": 0.4, "mean_velocity": 0.2, "track_id_count": 1,
16             "chunk_index": 0,
17         }
18
19
20     def _patched(fn):
21         """Stack the patches every pipeline test needs. fps/width probing is stubbed
22         so cv2 internals don't need mocking (covered separately).
23
24         Patches are applied innermost-first, and mock.patch injects the innermost
25         patch as the *first* positional arg ? so this order matches the signature
26         (mock_runner_cls, mock_probe, mock_dur, mock_extract, mock_provider_registry,
27         mock_env)."""
28         fn = patch("backend.pipeline.segmenters.tracknet_hybrid.TrackNetRunner")(fn)
29         fn = patch.object(TrackNetHybridSegmenter, "_probe_fps_width", return_value=(30.0,
30         1280.0))(fn)
31         fn = patch("backend.pipeline.segmenters.tracknet_hybrid.get_video_duration")(fn)
32         fn = patch("backend.pipeline.segmenters.tracknet_hybrid.extract_video_chunk")(fn)
33         fn = patch("backend.pipeline.segmenters.tracknet_hybrid.provider_registry")(fn)
34         fn = patch("os.environ.get")(fn)
35         return fn
36
37     def _make_runner(windows=None, csv="out.csv", healthy=True):
38         runner = MagicMock()
39         runner.healthcheck.return_value = (healthy, "TF 2.17 GPUs 1" if healthy else "no
GPU")
40         runner.run_predict.return_value = csv
41         runner.parse_trajectory_csv.return_value = [MagicMock(visible=True)]
42         runner.trajectory_to_action_windows.return_value = windows if windows is not None
43         else [_window()]
44         return runner
45
46     @_patched
47     def test_pipeline_happy_path(mock_runner_cls, mock_probe, mock_dur, mock_extract,
48                                 mock_provider_registry, mock_env):
49         mock_env.return_value = "dummy_key"
50         mock_dur.return_value = 30.0
51         mock_extract.return_value = True
52         mock_runner_cls.return_value = _make_runner()
53
54         provider = MagicMock()
55         provider.analyze_video.return_value = (
56             [{"start_time": 0.5, "end_time": 3.0, "shuttle_exchange_count": 4}], {})
57         mock_provider_registry.get.return_value = provider
58
59         db = MagicMock()
60         db.add_candidate_segment.return_value = 55
61         db.add_play_segment.return_value = 200
62
63         seg = TrackNetHybridSegmenter(db, {"indexing": {}})
64         res = seg.process_video("v1", "clip.mp4")
65
66         assert len(res) == 1
67         assert res[0]["id"] == 200
68         # Candidate persisted under the tracknet detector name...
69         _, kwargs = db.add_candidate_segment.call_args

```

```

70     assert kwargs["detector"] == "tracknet_hybrid"
71     # ...and the confirmed segment linked back to it.
72     db.link_candidate_to_segment.assert_called_once_with(55, 200)
73
74
75     @_patched
76     def test_pipeline_no_api_key_returns_empty(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
77                                             mock_provider_registry, mock_env):
78         mock_env.return_value = None
79         mock_dur.return_value = 30.0
80         mock_runner_cls.return_value = _make_runner()
81
82         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
83         assert seg.process_video("v1", "clip.mp4") == []
84
85
86     @_patched
87     def test_pipeline_healthcheck_failure_aborts(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
88                                             mock_provider_registry, mock_env):
89         mock_env.return_value = "dummy_key"
90         mock_dur.return_value = 30.0
91         mock_extract.return_value = True
92         mock_runner_cls.return_value = _make_runner(healthy=False)
93         mock_provider_registry.get.return_value = MagicMock()
94
95         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
96         res = seg.process_video("v1", "clip.mp4")
97         assert res == []
98         # Should bail before ever running inference.
99         mock_runner_cls.return_value.run_predict.assert_not_called()
100
101
102     @_patched
103     def test_pipeline_no_trajectory_aborts(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
104                                             mock_provider_registry, mock_env):
105         mock_env.return_value = "dummy_key"
106         mock_dur.return_value = 30.0
107         mock_runner_cls.return_value = _make_runner(csv=None) # predict produced nothing
108         mock_provider_registry.get.return_value = MagicMock()
109
110         seg = TrackNetHybridSegmenter(MagicMock(), {"indexing": {}})
111         assert seg.process_video("v1", "clip.mp4") == []
112
113
114     @_patched
115     def test_pipeline_store_candidates_disabled(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
116                                             mock_provider_registry, mock_env):
117         mock_env.return_value = "dummy_key"
118         mock_dur.return_value = 30.0
119         mock_extract.return_value = True
120         mock_runner_cls.return_value = _make_runner()
121
122         provider = MagicMock()
123         provider.analyze_video.return_value = ([{"start_time": 0.5, "end_time": 3.0}], {})
124         mock_provider_registry.get.return_value = provider
125
126         db = MagicMock()
127         db.add_play_segment.return_value = 200
128
129         seg = TrackNetHybridSegmenter(db, {"indexing": {"store_yolo_candidates": False}})
130         seg.process_video("v1", "clip.mp4")

```

```

131
132     db.add_candidate_segment.assert_not_called()
133     db.link_candidate_to_segment.assert_not_called()
134
135
136     @patched
137     def test_pipeline_drops_unparseable_segments(mock_runner_cls, mock_probe, mock_dur,
mock_extract,
138                                             mock_provider_registry, mock_env):
139         mock_env.return_value = "dummy_key"
140         mock_dur.return_value = 30.0
141         mock_extract.return_value = True
142         mock_runner_cls.return_value = _make_runner()
143
144         provider = MagicMock()
145         provider.analyze_video.return_value = (
146             [{"start_time": 0.5, "end_time": 3.0}, {"start_time": "N/A", "end_time": "??"}],
147             {})
148         mock_provider_registry.get.return_value = provider
149
150         db = MagicMock()
151         db.add_candidate_segment.return_value = 55
152         db.add_play_segment.return_value = 200
153
154         seg = TrackNetHybridSegmenter(db, {"indexing": {}})
155         res = seg.process_video("v1", "clip.mp4")
156         assert len(res) == 1 # the "N/A" segment is dropped
157
158     def test_probe_fps_width_fallbacks():
159         with patch("backend.pipeline.segmenters.tracknet_hybrid.cv2.VideoCapture") as
mock_cap:
160             bad = MagicMock()
161             bad.isOpened.return_value = False
162             bad.get.return_value = 0.0
163             mock_cap.return_value = bad
164             fps, width = TrackNetHybridSegmenter._probe_fps_width("missing.mp4")
165             assert fps == 30.0 and width == 1280.0
166
167
168     def test_segmenter_identity():
169         seg = TrackNetHybridSegmenter(MagicMock(), {})
170         assert seg.name == "tracknet_hybrid"
171         assert "TrackNet" in seg.description
172

```

26. user (2026-06-10T05:35:56.654Z)

```

1     """Tests for the validation registry, focused on the backward-compatible
2     `run_all_with_context` added for observability reporting."""
3     from backend.pipeline.validators.base import ValidationRegistry, ValidationResult,
VideoValidator
4
5
6     class _ProducerValidator(VideoValidator):
7         @property
8         def name(self):
9             return "producer"
10
11         @property
12         def description(self):
13             return "writes ffprobe_meta into the shared context"
14
15         def validate(self, video_path, config, context):
16             context["ffprobe_meta"] = {"fps": 59.94, "width": 1920, "height": 1080}

```

```

17         return ValidationResult(validator_name=self.name, passed=True, message="ok")
18
19
20     class _ConsumerValidator(VideoValidator):
21         @property
22         def name(self):
23             return "consumer"
24
25         @property
26         def description(self):
27             return "reads ffprobe_meta the producer cached"
28
29         def validate(self, video_path, config, context):
30             seen = "ffprobe_meta" in context
31             return ValidationResult(validator_name=self.name, passed=seen,
message=f"saw_meta={seen}")
32
33
34     def _registry():
35         r = ValidationRegistry()
36         r.register(_ProducerValidator())
37         r.register(_ConsumerValidator())
38         return r
39
40
41     def test_run_all_with_context_returns_results_and_context():
42         results, context = _registry().run_all_with_context("v.mp4", {})
43         assert [r.validator_name for r in results] == ["producer", "consumer"]
44         assert context["ffprobe_meta"]["fps"] == 59.94
45         assert results[1].passed is True # consumer saw the shared context
46
47
48     def test_run_all_is_unchanged_and_returns_only_results():
49         results = _registry().run_all("v.mp4", {})
50         assert isinstance(results, list)
51         assert all(isinstance(r, ValidationResult) for r in results)
52         assert [r.validator_name for r in results] == ["producer", "consumer"]
53
54
55     def test_validator_exception_is_captured_not_raised():
56         class _Boom(VideoValidator):
57             @property
58             def name(self):
59                 return "boom"
60
61             @property
62             def description(self):
63                 return "raises"
64
65             def validate(self, video_path, config, context):
66                 raise RuntimeError("explode")
67
68         r = ValidationRegistry()
69         r.register(_Boom())
70         results, _ = r.run_all_with_context("v.mp4", {})
71         assert results[0].passed is False
72         assert "explode" in results[0].message
73

```

27. user (2026-06-10T05:36:18.449Z)

```

cv2 4.13.0 numpy 2.4.6
tests/test_eval_classifier.py:1:import numpy as np
tests/test_hit_detector.py:2:import numpy as np
tests/test_rally_seq_proto.py:9:import numpy as np

```

```
tests/test_tracknet_hybrid.py:5:import cv2
tests/test_wasb_cache.py:11:import numpy as np
tests/test_yolo_hybrid.py:8:    import cv2
tests/test_yolo_hybrid.py:279:    import numpy as np
```

28. user (2026-06-10T05:36:18.680Z)

```
1c1
< """WASB-based hybrid play segmenter.
---
> """TrackNet-based hybrid play segmenter.
3,11c3,12
< Identical three-phase shape to TrackNetHybridSegmenter, but Phase 2 uses **WASB**
< (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via WSL2
< (backend/detectors/wasb_runner.py) instead of TrackNetV4. WASB ships pretrained
< badminton weights and runs today, so this is the live shuttle-motion substrate.
< Phase 3 (AI semantic boundaries) and Phase 4 (DB) are shared with the other hybrids.
<
< For the per-video *calibration* goal we mainly use Phase 2 (trajectory -> candidate
< windows); the windowing thresholds (velocity_thresh / merge_gap / min_window_duration)
< are what get tuned against golden rally labels.
---
> Same three-phase shape as HybridYoloSegmenter, but Phase 2 (candidate
> generation) uses a TrackNetV4 *shuttle* trajectory instead of YOLO *person*
> motion. TrackNet runs in WSL2 (see backend/detectors/tracknet_runner.py and
> docs/DECISIONS.md ADR-001); the shuttle trajectory is converted into candidate
> rally windows, which are then handed to the AI provider for semantic boundary
> confirmation (Phase 3) and written to the DB (Phase 4).
>
> Rationale: a tiny, fast, motion-blurred shuttle is exactly what general
> detectors miss; a purpose-built trajectory model localises it far more
> precisely, giving cleaner candidate rallies than person-velocity heuristics.
12a14
>
25c27
< from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
---
> from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
38c40
< class WasbHybridSegmenter(BasePlaySegmenter):
---
> class TrackNetHybridSegmenter(BasePlaySegmenter):
41c43
<     return "wasb_hybrid"
---
>     return "tracknet_hybrid"
45,46c47,48
<     return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
<         "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
---
>     return ("Two-stage pipeline: TrackNetV4 shuttle-trajectory detection (WSL2) for
precise "
>         "candidate rallies + AI provider for high-precision semantic boundaries.")
49a52
>     """Return (fps, frame_width) for a video, with sensible fallbacks."""
57a61
>     # --- Sport profile + AI provider wiring (identical to the other segmenters) ---
72c76,80
<         logger.error(f"{api_key_env_var} not set; cannot run the {provider_name}
handoff.")
---
>         logger.error(
>             f"{api_key_env_var} environment variable is not set! "
```

```

>         f"To run the {provider_name} handoff, set your API key. "
>         f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
>     )
86,87c94,96
<     wasb_cfg = WasbConfig.from_indexing_cfg(idx_cfg)
<     runner: DetectorRunner = WasbRunner(wasb_cfg)
---
>     # --- TrackNet runner config + thresholds ---
>     tn_cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
>     runner: DetectorRunner = TrackNetRunner(tn_cfg)
89,90c98,99
<     w = idx_cfg.get("wasb", {})
<     velocity_thresh = w.get("velocity_threshold", 0.02)
---
>     tn = idx_cfg.get("tracknet", {})
>     velocity_thresh = tn.get("velocity_threshold", 0.02) # normalised shuttle
displacement/sec
99c108
<         "detector": "wasb_hybrid",
---
>         "detector": "tracknet_hybrid",
103c112,113
<         "weights_path": wasb_cfg.weights_path,
---
>         "weights_path": tn_cfg.weights_path,
>         "queue_length": tn_cfg.queue_length,
105a116

```

29. user (2026-06-10T05:36:33.156Z)

```

torch 2.6.0+cu124
191
>         logger.error("TrackNet produced no trajectory; aborting.")
129c142
<         logger.info("Phase 2 (WASB) completed in %.1fs. Candidate windows: %d",
---
>         logger.info("Phase 2 (TrackNet) completed in %.1fs. Candidate windows: %d",
133,136c146,151
<         for cw in all_candidate_windows:
<             logger.info(f"[WASB rally] {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} "
<                         f"({cw['end'] - cw['start']:.1f}s) |
frames={cw['active_frame_count']} "
<                         f"peak_v={cw['peak_velocity']:.3f}
mean_v={cw['mean_velocity']:.3f}")
---
>         for w in all_candidate_windows:
>             logger.info(
>                 f"[TrackNet rally] {_fmt_ts(w['start'])}-{_fmt_ts(w['end'])} "
>                 f"({w['end'] - w['start']:.1f}s) | frames={w['active_frame_count']} "
>                 f"peak_v={w['peak_velocity']:.3f} mean_v={w['mean_velocity']:.3f}"
>             )
137a153
>         # Persist raw candidates for the fine-tuning dataset (same table as YOLO).
140c156
<         for i, cw in enumerate(all_candidate_windows):
---
>         for i, w in enumerate(all_candidate_windows):
142,143c158,161
<             "peak_velocity": cw["peak_velocity"], "mean_velocity":
cw["mean_velocity"],
<             "active_frame_count": cw["active_frame_count"], "track_id_count":
cw["track_id_count"],
---
>             "peak_velocity": w["peak_velocity"],
>             "mean_velocity": w["mean_velocity"],

```

```

>         "active_frame_count": w["active_frame_count"],
>         "track_id_count": w["track_id_count"],
144a163
>         confidence = min(1.0, w["peak_velocity"])
146,147c165,167
<         video_id, detector="wasb_hybrid", start_time=cw["start"],
end_time=cw["end"],
<         chunk_index=cw["chunk_index"], confidence=min(1.0, cw["peak_velocity"]),
---
>         video_id, detector="tracknet_hybrid",
>         start_time=w["start"], end_time=w["end"],
>         chunk_index=w["chunk_index"], confidence=confidence,
154c174
<         # --- Phase 3: AI provider handoff (shared pattern) ---
---
>         # --- Phase 3: AI provider handoff (identical pattern to HybridYoloSegmenter) ---
187,188c207,208
<         futures = [executor.submit(process_candidate_window, wi, cw)
<             for wi, cw in enumerate(all_candidate_windows)]
---
>         futures = [executor.submit(process_candidate_window, wi, w)
>             for wi, w in enumerate(all_candidate_windows)]
190a211
>
197c218
<         # --- Phase 4: DB population (shared pattern) ---
---
>         # --- Phase 4: DB population (identical pattern to HybridYoloSegmenter) ---
205,207c226
<         try:
<             shot_count = int(shot_count)
<         except (ValueError, TypeError):
---
>         if shot_count is None:
209,211c228,237
<         metadata = {"shot_count": shot_count,
<             "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
<             "detector": f"wasb-hybrid-{model_name}"}
---
>         else:
>             try:
>                 shot_count = int(shot_count)
>             except (ValueError, TypeError):
>                 shot_count = max(3, int(dur / 0.8))
>         metadata = {
>             "shot_count": shot_count,
>             "shot_count_confidence": segment.get("shot_count_confidence", "Unknown"),
>             "detector": f"tracknet-hybrid-{model_name}",
>         }
217,218c243,246
<         segments_data.append({"id": segment_id, "start_time": start, "end_time":
end,
<             "duration": dur, "metadata": metadata})
---
>         segments_data.append({
>             "id": segment_id, "start_time": start, "end_time": end,
>             "duration": dur, "metadata": metadata,
>         })
225c253,254
< segmenter_registry.register(WasbHybridSegmenter)
---
> # Register the segmenter
> segmenter_registry.register(TrackNetHybridSegmenter)

```


30. user (2026-06-10T05:36:45.252Z)

```
195         logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
196 len(ai_segments))
197
198     # --- Phase 4: DB population (shared pattern) ---
199     segments_data = []
200     for segment in ai_segments:
201         try:
202             start = float(segment["start_time"])
203             end = float(segment["end_time"])
204             dur = end - start
205             shot_count = segment.get("shuttle_exchange_count")
206             try:
207                 shot_count = int(shot_count)
208             except (ValueError, TypeError):
209                 shot_count = max(3, int(dur / 0.8))
210             metadata = {"shot_count": shot_count,
211                         "shot_count_confidence":
212 segment.get("shot_count_confidence", "Unknown"),
213                         "detector": f"wasb-hybrid-{model_name}"}
214             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
215 metadata)
216             if segment_id:
217                 candidate_id = segment.get("_candidate_id")
218                 if candidate_id is not None:
219                     self.db.link_candidate_to_segment(candidate_id, segment_id)
220                 segments_data.append({"id": segment_id, "start_time": start,
221 "end_time": end,
222                                     "duration": dur, "metadata": metadata})
223         except Exception as e:
224             logger.warning("Skipping invalid segment struct: %s", e)
225
226     return segments_data
227
228 segmenter_registry.register(WasbHybridSegmenter)
```

31. user (2026-06-10T05:36:45.624Z)

```
1     """Resumable-cache layer of wasb_infer (pure-Python parts; no WSL/GPU).
2
3     Covers the reboot-resume guarantees: per-window incremental cache, cross-window
4     detection accumulation on replay, crash-truncation recovery, manifest
5     compatibility/invalidation, and atomic trajectory writes.
6     """
7     import json
8     import os
9     import os.path as osp
10
11     import numpy as np
12
13     from backend.pipeline.detectors import wasb_infer as wi
14
15
16     def _win(idx, frames):
17         """A window record: frames = [(fid, [[x,y,score,scale], ...]), ...]."""
18         return {"w": idx, "f": [[fid, dets] for fid, dets in frames]}
19
20
21     def _write_windows(cache_dir, windows):
22         det_jsonl, _ = wi._cache_paths(cache_dir)
23         with open(det_jsonl, "a") as f:
24             wi._append_windows(f, windows)
25
```

```

26
27 # ----- #
28 # detection (de)serialization
29 # ----- #
30 def test_det_plain_and_back_roundtrip():
31     det = {"xy": np.array([439.0, 64.5]), "score": np.float32(0.87), "scale": 0}
32     plain = wi._det_plain(det)
33     assert plain == [439.0, 64.5, float(np.float32(0.87)), 0]
34
35     back = wi._dets_to_tracker([plain])
36     assert len(back) == 1
37     assert isinstance(back[0]["xy"], np.ndarray) # tracker does vector math on xy
38     assert back[0]["xy"].tolist() == [439.0, 64.5]
39     assert back[0]["scale"] == 0
40
41
42 # ----- #
43 # cache replay + resume index
44 # ----- #
45 def test_replay_accumulates_detections_across_windows(tmp_path):
46     cache = str(tmp_path / "c")
47     os.makedirs(cache)
48     # frame 1 appears in BOTH windows -> its candidates must accumulate in window order;
49     # frame 2 is empty in window 0 but detected in window 1.
50     _write_windows(cache, [
51         _win(0, [(0, [[10, 20, 0.9, 0]]), (1, [[11, 21, 0.8, 0]]), (2, [])]),
52         _win(1, [(1, [[12, 22, 0.7, 0]]), (2, [[13, 23, 0.6, 0]]), (3, [])]),
53     ])
54     windows_done, det = wi.load_and_clean_cache(cache)
55     assert windows_done == 2
56     assert det[0] == [[10, 20, 0.9, 0]]
57     assert det[1] == [[11, 21, 0.8, 0], [12, 22, 0.7, 0]] # window-0 then window-1
58     assert det[2] == [[13, 23, 0.6, 0]]
59     assert det[3] == [] # seen but never detected
60
61
62 def test_resume_index_equals_window_count(tmp_path):
63     cache = str(tmp_path / "c")
64     os.makedirs(cache)
65     _write_windows(cache, [_win(i, [(i, [[i, i, 0.5, 0]])]) for i in range(5)])
66     windows_done, _ = wi.load_and_clean_cache(cache)
67     assert windows_done == 5
68
69
70 def test_empty_cache(tmp_path):
71     cache = str(tmp_path / "c")
72     os.makedirs(cache)
73     windows_done, det = wi.load_and_clean_cache(cache)
74     assert windows_done == 0 and dict(det) == {}
75
76
77 # ----- #
78 # crash robustness
79 # ----- #
80 def test_truncated_trailing_line_recovered_and_cleaned(tmp_path):
81     cache = str(tmp_path / "c")
82     os.makedirs(cache)
83     _write_windows(cache, [_win(0, [(0, [[1, 1, 0.9, 0]])]),
84                             _win(1, [(1, [[2, 2, 0.8, 0]])])])
85     det_jsonl, _ = wi._cache_paths(cache)
86     with open(det_jsonl, "a") as f: # simulate a crash mid-flush
87         f.write({'w': 2, "f": [[2, [[3, 3, 0]]]})
88     windows_done, det = wi.load_and_clean_cache(cache)
89     assert windows_done == 2 # only the two clean windows

```

survive

```

90     assert det[1] == [[2, 2, 0.8, 0]]
91     # file is rewritten to the clean prefix -> appending window 2 stays contiguous
92     with open(det_jsonl) as f:
93         lines = [json.loads(l) for l in f if l.strip()]
94     assert [o["w"] for o in lines] == [0, 1]
95     _write_windows(cache, [_win(2, [(2, [[3, 3, 0.7, 0]]))])
96     assert wi.load_and_clean_cache(cache)[0] == 3
97
98
99 def test_noncontiguous_window_index_stops_replay(tmp_path):
100     cache = str(tmp_path / "c")
101     os.makedirs(cache)
102     _write_windows(cache, [_win(0, [(0, [[1, 1, 0.9, 0]]))],
103                          _win(5, [(5, [[2, 2, 0.8, 0]]))]) # gap -> stop after w=0
104     windows_done, det = wi.load_and_clean_cache(cache)
105     assert windows_done == 1
106     assert 5 not in det
107
108
109 # ----- #
110 # manifest: atomic write + compatibility / invalidation
111 # ----- #
112 def test_manifest_roundtrip_and_compat(tmp_path):
113     cache = str(tmp_path / "c")
114     os.makedirs(cache)
115     m = {
116         "version": wi.CACHE_VERSION,
117         "frames_dir": osp.abspath("/x/frames"),
118         "weights": "w.pth.tar",
119         "sport": "badminton",
120         "frames_in": 3,
121         "frames_total": 100,
122         "windows_total": 98,
123         "windows_done": 42,
124     }
125     wi.write_manifest(cache, m)
126     assert not osp.exists(osp.join(cache, wi._MANIFEST_FILE + ".tmp")) # atomic, no
tmp left
127     assert wi.read_manifest(cache) == m
128
129     base = dict(frames_dir="/x/frames", weights="w.pth.tar", sport="badminton",
130                frames_in=3, frames_total=100)
131     assert wi.manifest_compatible(m, **base)
132     assert not wi.manifest_compatible(m, **{**base, "frames_total": 101}) # more
frames
133     assert not wi.manifest_compatible(m, **{**base, "weights": "other.pth"})
134     assert not wi.manifest_compatible(m, **{**base, "sport": "tennis"})
135     assert not wi.manifest_compatible({}, **base) # no cache
136     assert not wi.manifest_compatible(**m, "version": 999, **base) # schema bump
137
138
139 def test_read_manifest_missing_returns_none(tmp_path):
140     assert wi.read_manifest(str(tmp_path)) is None
141
142
143 def test_reset_cache_removes_files(tmp_path):
144     cache = str(tmp_path / "c")
145     os.makedirs(cache)
146     _write_windows(cache, [_win(0, [(0, [[1, 1, 0.9, 0]]))])
147     wi.write_manifest(cache, {"version": wi.CACHE_VERSION})
148     wi.reset_cache(cache)
149     det_jsonl, mpath = wi._cache_paths(cache)
150     assert not osp.exists(det_jsonl) and not osp.exists(mpath)
151
152

```

```

153 # ----- #
154 # misc helpers
155 # ----- #
156 def test_default_cache_dir_is_next_to_output():
157     out = osp.join("some", "dir", "sample_long_traj.csv")
158     cd = wi.default_cache_dir(out)
159     assert osp.basename(cd) == "sample_long_traj_wasbcache"
160     assert osp.dirname(cd) == osp.dirname(osp.abspath(out))
161
162
163 def test_atomic_write_trajectory(tmp_path):
164     out = str(tmp_path / "out" / "traj.csv")
165     wi._atomic_write_trajectory(out, [(0, 1, 439.0, 64.0), (1, 0, -1, -1)])
166     assert not osp.exists(out + ".tmp")
167     with open(out) as f:
168         text = f.read().splitlines()
169     assert text[0] == "Frame,Visibility,X,Y"
170     assert text[1] == "0,1,439.0,64.0"
171     assert text[2] == "1,0,-1,-1"
172

```

32. user (2026-06-10T05:37:03.342Z)

--- API Endpoints ---

```

@app.get("/api/config")
def get_config():
    return global_config

@app.post("/api/process")
def process_video(req: ProcessRequest):
    if not os.path.exists(req.video_path):
        raise HTTPException(status_code=404, detail=f"Video file not found at
'{req.video_path}'")

    video_id = compute_video_id(req.video_path)
    was_reingest = db_instance.get_video(video_id) is not None

    # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the report)
    logger.info(f"Running validations for {req.video_path}...")
    val_results, val_context = registry.run_all_with_context(req.video_path, global_config)
    failures = [r for r in val_results if not r.passed]

    # typed AppConfig: attribute access for the default segmenter (with safe fallback)
    default_seg = getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
    seg_name = req.segmenter or default_seg
    segmenter_actual = None
    segmentation_ran = False
    response: Optional[Dict[str, Any]] = None
    try:
        if failures:
            db_instance.add_video(video_id, req.video_path, "failed", [r.dict() for r in
val_results])
            response = {
                "video_id": video_id,
                "status": "failed",
                "message": "Video failed validation checks.",
                "results": [r.dict() for r in val_results],
            }
        return response

    # 2. Run segmenter & indexer
    print(f"[API] Video passed checks. Segmenting and indexing...")
    db_instance.add_video(video_id, req.video_path, "processed", [r.dict() for r in

```

```

val_results])

    segmenter_cls = segmenter_registry.get(seg_name)
    if not segmenter_cls:
        available = list(segmenter_registry.list_available().keys())
        raise HTTPException(
            status_code=400,
            detail=f"Segmenter '{seg_name}' not found. Available segmenters: {available}"
        )

    print(f"[API] Using segmenter: {seg_name}")
    segmenter_actual = seg_name
    segmenter = segmenter_cls(db_instance, global_config)
    result = segmenter.process_video(video_id, req.video_path, req.player_pool,
req.max_frames)
    segmentation_ran = True

    if isinstance(result, Failure):
        print(f"[API] Segmenter failed: {result.message}")
        response = {
            "video_id": video_id,
            "status": "failed",
            "message": f"Segmenter '{seg_name}' failed: {result.message}",
            "results": [r.dict() for r in val_results],
            "play_segments": [],
        }
        return response

    segments = result.value if hasattr(result, "value") else result

    response = {
        "video_id": video_id,
        "status": "success",
        "message": f"Successfully indexed {len(segments)} play segments using '{seg_name}'
segmenter.",
        "results": [r.dict() for r in val_results],
        "play_segments": segments,
    }
    return response
finally:
    # Always emit the per-video observability report (next to the video, or to
    # report.output_dir). Never raises. Surface the paths in the response.
    paths = emit_indexer_report(
        db=db_instance, video_id=video_id, video_path=req.video_path, config=global_config,
        val_results=val_results, val_context=val_context,
        segmenter_requested=req.segmenter, segmenter_actual=segmenter_actual,
        was_reingest=was_reingest, segmentation_ran=segmentation_ran,
    )
    if paths and isinstance(response, dict):
        response["reports"] = paths
26:     "default_segmenter": "gemini",

```

33. user (2026-06-10T05:37:05.703Z)

```

17:from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
218:def test_wasb_runner_implements_detector_runner_abc():
219:     """WasbRunner must also satisfy the DetectorRunner contract."""
220:     w = WasbRunner(WasbConfig())
===
43:     def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "WasbConfig":
58:class WasbRunner(DetectorRunner): # DetectorRunner ABC (finish item 6 / new item 1)
59:     def __init__(self, cfg: WasbConfig):
62:     def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
69:     def healthcheck(self) -> Tuple[bool, str]:
85:     def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:

```

```

99:     def _stage_infer_script(self) -> bool:
106:     def run_predict(self, video_win_path: str, output_win_dir: str) -> Optional[str]:
    150 backend/pipeline/detectors/wasb_runner.py
    299 backend/pipeline/detectors/tracknet_runner.py
    410 backend/pipeline/detectors/wasb_infer.py
    859 total

```

34. user (2026-06-10T05:37:18.458Z)

```

1     import pytest
2     from backend.pipeline.segmenters.base import BasePlaySegmenter, SegmenterRegistry
3     from backend.results import Success, Failure, Result # test the new contract
4
5     class DummySegmenter(BasePlaySegmenter):
6         @property
7         def name(self) -> str:
8             return "dummy"
9
10        @property
11        def description(self) -> str:
12            return "Dummy segmenter for testing."
13
14        def process_video(self, video_id, video_path, player_pool=None, max_frames=None) ->
Result:
15            # Demonstrate both success and failure paths for the contract
16            if video_path == "fail":
17                return Failure(message="Simulated failure for test", is_recoverable=True)
18            return Success(value=[{"start_time": 0.0, "end_time": 1.0}])
19
20    def test_registry():
21        registry = SegmenterRegistry()
22        registry.register(DummySegmenter)
23
24        available = registry.list_available()
25        assert "dummy" in available
26        assert available["dummy"] == "Dummy segmenter for testing."
27
28        cls = registry.get("dummy")
29        assert cls == DummySegmenter
30
31        inst = cls(None, {})
32        assert inst.name == "dummy"
33        success = inst.process_video("v1", "path")
34        assert isinstance(success, Success)
35        assert success.is_success() is True
36        assert success.value == [{"start_time": 0.0, "end_time": 1.0}]
37        assert success.unwrap() == [{"start_time": 0.0, "end_time": 1.0}]
38
39        failure = inst.process_video("v1", "fail")
40        assert isinstance(failure, Failure)
41        assert failure.is_success() is False
42        assert failure.is_recoverable is True
43        assert "Simulated failure" in failure.message
44        try:
45            failure.unwrap()
46            assert False, "Should have raised"
47        except RuntimeError as e:
48            assert "Simulated failure" in str(e)
49

```

35. user (2026-06-10T05:37:18.611Z)

```

2:parsing, trajectory -> action-window conversion). The WSL subprocess invocation
6:import subprocess
8:from unittest.mock import patch

```

```

14:     to_wsl_mnt_path,
21:     return subprocess.CompletedProcess(args=[], returncode=returncode,
25: def test_to_wsl_mnt_path_windows():
26:     assert to_wsl_mnt_path(r"C:\Users\avidu\v.mp4") == "/mnt/c/Users/avidu/v.mp4"
27:     assert to_wsl_mnt_path(r"D:\clips\m.mp4") == "/mnt/d/clips/m.mp4"
30: def test_to_wsl_mnt_path_passthrough():
32:     assert to_wsl_mnt_path("/home/avidullu/clips/v.mp4") == "/home/avidullu/clips/v.mp4"
33:     assert to_wsl_mnt_path("~/clips/v.mp4") == "~/clips/v.mp4"
36: def test_config_from_indexing_cfg():
38:     "wsl_distro": "Ubuntu-22.04", "conda_env": "TN", "weights_path": "/w/m.h5",
39:     "queue_length": 7, "stage_in_wsl": False,
45:     assert cfg.stage_in_wsl is False
48: def test_config_defaults_when_absent():
55: def test_parse_trajectory_csv(tmp_path):
71: def test_trajectory_to_action_windows_detects_moving_shuttle():
90: def test_trajectory_to_action_windows_ignores_stationary_shuttle():
100: def test_trajectory_to_action_windows_splits_on_gap():
114: # ----- WSL subprocess glue
120: def test_healthcheck_ok():
122:     with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="TF 2.17.0 GPUs 1")):
128: def test_healthcheck_failure_returns_message():
130:     with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="ModuleNotFoundError")):
136: def test_healthcheck_no_wsl_binary():
138:     with patch.object(r, "_wsl_bash", side_effect=FileNotFoundError()):
141:     assert "wsl" in msg.lower()
144: def test_run_predict_requires_weights():
149: def test_run_predict_rejects_bad_extension(tmp_path):
154: def test_run_predict_success(tmp_path):
155:     r = TrackNetRunner(TrackNetConfig(weights_path="/w/model.h5", stage_in_wsl=False))
159:     with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="Prediction complete.")):
165: def test_run_predict_failure_returns_none(tmp_path):
166:     r = TrackNetRunner(TrackNetConfig(weights_path="/w/model.h5", stage_in_wsl=False))
167:     with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="CUDA error")):
171: def test_run_predict_missing_output_csv(tmp_path):
173:     r = TrackNetRunner(TrackNetConfig(weights_path="/w/model.h5", stage_in_wsl=False))
174:     with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="done")):
178: def test_stage_video_success():
=== compile endpoint ===
@app.post("/api/compile")
def compile_highlights(req: CompileRequest):
    video = db_instance.get_video(req.video_id)
    if not video:
        raise HTTPException(status_code=404, detail="Indexed video record not found.")

    # Retrieve matching play segments from db
    all_segments = db_instance.query_play_segments(req.video_id, {})
    matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]

    if not matched_segments:
        raise HTTPException(status_code=400, detail="No matching play segments found to
stitch.")

    # Extract start/end time pairs
    time_pairs = [(s["start_time"], s["end_time"]) for s in matched_segments]

    # Run compiler
    output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output") or
"output"
    compiler = VideoCompiler(output_dir)
    try:
        out_path = compiler.compile_highlights(video["filepath"], time_pairs,
req.output_filename)
        return {"status": "success", "filepath": out_path}
    except Exception as e:
        raise HTTPException(status_code=500, detail=f"Highlight stitching failed: {str(e)}")

```

--- CLI Debug Utility & Orchestration ---

36. user (2026-06-10T05:37:29.576Z)

```
tests/test_providers.py:2:from backend.providers import provider_registry
tests/test_providers.py:3:from backend.providers.base import AIVideoProvider
tests/test_providers.py:4:from backend.providers.gemini import GeminiProvider
tests/test_providers.py:5-
tests/test_providers.py:6:def test_provider_registry():
tests/test_providers.py:7:    available = provider_registry.list_available()
tests/test_providers.py:8:    assert "gemini" in available
tests/test_providers.py:9-
tests/test_providers.py:10:    # Retrieve provider instance
tests/test_providers.py:11:    provider = provider_registry.get("gemini", api_key="dummy",
model_name="gemini-2.5-flash")
tests/test_providers.py:12:    assert isinstance(provider, GeminiProvider)
--
tests/test_sport_profiles.py:2:from backend.sports import sport_registry
tests/test_sport_profiles.py:3:from backend.sports.base import SportProfile
tests/test_sport_profiles.py:4:from backend.sports.badminton import BadmintonSportProfile
tests/test_sport_profiles.py:5-
tests/test_sport_profiles.py:6:def test_sport_registry():
tests/test_sport_profiles.py:7:    # Verify badminton is registered
tests/test_sport_profiles.py:8:    available = sport_registry.list_available()
tests/test_sport_profiles.py:9:    assert "badminton" in available
tests/test_sport_profiles.py:10-
tests/test_sport_profiles.py:11:    # Retrieve badminton profile
tests/test_sport_profiles.py:12:    badminton = sport_registry.get("badminton")
tests/test_sport_profiles.py:13:    assert isinstance(badminton, BadmintonSportProfile)
tests/test_sport_profiles.py:14:    assert badminton.name == "badminton"
tests/test_sport_profiles.py:15-
=== annotation reg ===
tests/test_annotation_providers.py:3:from backend.annotations import annotation_registry,
AnnotationProvider
tests/test_annotation_providers.py:17:    assert "file" in annotation_registry.list_available()
tests/test_annotation_providers.py:18:    p = annotation_registry.get("file")
tests/test_annotation_providers.py:25:    annotation_registry.get("does-not-exist")
tests/test_annotation_providers.py:29:    p = annotation_registry.get("file",
csv_dir=str(tmp_path))
tests/test_automated_provider.py:7:from backend.annotations import annotation_registry
tests/test_automated_provider.py:22:    assert "auto" in annotation_registry.list_available()
tests/test_automated_provider.py:23:    p = annotation_registry.get("auto")
tests/test_automated_provider.py:29:    p = annotation_registry.get("auto",
labeler=_fake_labeler, lineage="fake")
tests/test_automated_provider.py:58:    p = annotation_registry.get("auto") # no labeler
injected
=== yolo cv2 ===
import pytest
from unittest.mock import MagicMock, patch
from backend.pipeline.segmenters.yolo_hybrid import HybridYoloSegmenter

def _make_cap_mock(num_frames=300):
    """A cv2.VideoCapture mock yielding `num_frames` readable frames at 30fps."""
    import cv2
    cap_mock = MagicMock()
    cap_mock.isOpened.return_value = True

    def mock_get_prop(prop):
        if prop == cv2.CAP_PROP_FPS:
            return 30.0
        if prop == cv2.CAP_PROP_FRAME_WIDTH:
            return 1000.0
        return 0.0
```



```

cap_mock.get.side_effect = mock_get_prop
cap_mock.read.side_effect = [(True, "frame")] * num_frames + [(False, None)]
return cap_mock

def _make_detections(num_frames=300, step=5, track_id=1):
    """Per-frame return values for a mocked _detect_and_track: one steadily-moving
    person. Each entry is the list of (track_id, bbox) for that frame, where the box
    advances `step` px/frame so its normalised velocity clears the test thresholds.
    """
    out = []
    for i in range(num_frames):
        x = i * step
        out.append([(track_id, (float(x), 0.0, float(x + 10), 10.0))])
    return out

def _mock_stage1(segmenter, num_frames=300, step=5):
    """Wire a segmenter's Stage-1 detection+tracking to deterministic fakes so tests
    never need torch model inference, GPU, or a real supervision tracker."""
    segmenter.model = MagicMock() # short-circuits _lazy_load_model
    segmenter._make_tracker = MagicMock(return_value=MagicMock())

```

37. user (2026-06-10T05:37:37.928Z)

```

1      """Basic tests for the typed configuration package.
2
3      These ensure:
4      - Defaults are sensible and come from the models.
5      - File overrides work.
6      - Old config files with extra keys are tolerated.
7      - save_default_config produces loadable output.
8      """
9
10     import json
11     import tempfile
12     import os
13
14     import pytest
15
16     from backend.config import load_config, save_default_config, AppConfig
17
18
19     def test_load_default_config():
20         # Force load without the project's config.json so we test the model built-in
21         defaults
22         cfg = load_config("/nonexistent/path/that/does/not/exist/12345/config.json")
23         assert isinstance(cfg, AppConfig)
24         assert cfg.sport == "badminton"
25         assert cfg.indexing.default_segmenter == "yolo_hybrid"
26         assert cfg.validation.min_blur_threshold == 20.0 # updated default from review
27         assert cfg.indexing.ai_max_workers == 5
28
29     def test_file_override():
30         with tempfile.TemporaryDirectory() as td:
31             p = os.path.join(td, "override.json")
32             data = {
33                 "indexing": {
34                     "default_segmenter": "gemini",
35                     "ai_max_workers": 3,
36                 },
37                 "validation": {
38                     "min_blur_threshold": 42.0,
39                 },

```

```

40         }
41         with open(p, "w") as f:
42             json.dump(data, f)
43
44         cfg = load_config(p)
45         assert cfg.indexing.default_segmenter == "gemini"
46         assert cfg.indexing.ai_max_workers == 3
47         assert cfg.validation.min_blur_threshold == 42.0
48         # Unspecified keys fall back to defaults
49         assert cfg.sport == "badminton"
50
51
52     def test_extra_keys_tolerated():
53         """Old or future keys in config.json must not break loading."""
54         with tempfile.TemporaryDirectory() as td:
55             p = os.path.join(td, "with_extra.json")
56             data = {
57                 "sport": "badminton",
58                 "indexing": {"default_segmenter": "yolo_hybrid"},
59                 "some_future_key": "ignored_for_now",
60                 "validation": {"relevance": {"court_pixels_min_ratio": 0.1}},
61             }
62             with open(p, "w") as f:
63                 json.dump(data, f)
64
65             cfg = load_config(p)
66             assert cfg.indexing.default_segmenter == "yolo_hybrid"
67             assert cfg.validation.relevance.court_pixels_min_ratio == 0.1
68
69
70     def test_save_default_config_roundtrip():
71         with tempfile.TemporaryDirectory() as td:
72             p = os.path.join(td, "saved.json")
73             saved = save_default_config(p)
74             assert str(saved) == p
75             assert os.path.exists(p)
76
77             cfg = load_config(p)
78             assert cfg.indexing.default_segmenter == "yolo_hybrid"
79             assert cfg.validation.min_blur_threshold == 20.0
80
81
82     def test_model_is_the_source_of_truth():
83         """Changing a default in the model should be reflected without touching json."""
84         # This is a documentation test more than anything.
85         cfg = AppConfig()
86         assert cfg.indexing.tracknet.velocity_threshold == 0.02
87
88
89     def test_output_dir_field_default_and_override():
90         """output_dir now lives on OutputConfig (was a write-only runtime hack), so
91         /api/compile and the CLI resolve the same value. Default + file override + the
92         runtime mutation that backend/main.py::main performs for --output-dir."""
93         cfg = AppConfig()
94         assert cfg.output.output_dir == "output" # built-in default
95
96         # The CLI override path: main() assigns args.output_dir onto the typed model.
97         cfg.output.output_dir = "/tmp/custom_out"
98         assert cfg.output.output_dir == "/tmp/custom_out"
99
100         # File override is honored on load.
101         with tempfile.TemporaryDirectory() as td:
102             p = os.path.join(td, "out_override.json")
103             with open(p, "w") as f:
104                 json.dump({"output": {"output_dir": "reels", "db_name": "x.db"}}, f)

```

```

105         loaded = load_config(p)
106         assert loaded.output.output_dir == "reels"
107         assert loaded.output.db_name == "x.db"
108
109
110     def test_get_returns_dict_for_nested_sections_legacy_compat():
111         """Regression: legacy consumers (validators/segmenters) do
112         ``config.get("validation", {}).get("min_resolution_width", ...)``. The dict-compat
113         .get must return nested SECTIONS as plain dicts (not Pydantic models), or that
114         chained access raises 'X object has no attribute get' on every real run."""
115         cfg = AppConfig()
116         validation = cfg.get("validation", {})
117         assert isinstance(validation, dict)
118         assert validation.get("min_resolution_width") == 1280
119         # nested-of-nested is also dict-accessible
120         assert isinstance(validation.get("relevance"), dict)
121         assert validation["relevance"].get("court_pixels_min_ratio") == 0.05
122         assert cfg.get("indexing", {}).get("default_segmenter") == "yolo_hybrid"
123         # scalar leaves are returned unchanged; a present scalar ignores the default arg;
124         # a missing key falls back to the default
125         assert cfg.get("ai_provider") == "gemini"
126         assert cfg.get("ai_provider", "ignored-default") == "gemini"
127         assert cfg.get("does_not_exist", "fallback") == "fallback"
128         # leaf keys inside any section resolve via the dynamic section scan (not a hardcoded
129         # list), so a future top-level section's keys would be discoverable too
130         assert cfg.get("min_resolution_width") == 1280 # lives inside `validation`
131         assert cfg.get("begin_padding") == 0.5 # lives inside `stitching`
132

```

38. user (2026-06-10T05:37:38.649Z)

```

30         parts.reverse()
31         total = 0.0
32         for i, p in enumerate(parts):
33             try:
34                 total += float(p) * (60 ** i)
35             except ValueError:
36                 pass
37         return total
38     try:
39         return float(val)
40     except ValueError:
41         logger.warning(f"Could not parse timestamp value '{val}' as a number.
Defaulting to 0.0.")
42         return 0.0
43         logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
'{val}'. Defaulting to 0.0.")
44         return 0.0
45
46     def _get_video_duration(self, video_path: str) -> float:
47         """Probes the video file to get its total duration in seconds.
48         Returns 0.0 if it cannot be determined."""
49         try:
50             result = subprocess.run(
51                 [
52                     "ffprobe", "-v", "error",
53                     "-show_entries", "format=duration",
54                     "-of", "default=noprint_wrappers=1:nokey=1",
55                     video_path
56                 ],
57                 capture_output=True, text=True, check=True
58             )
59             return float(result.stdout.strip())
60         except Exception as e:
61             logger.warning(f"Could not determine video duration via ffprobe: {e}")

```

```

62         return 0.0
63
64     def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float,
65 float]], output_filename: str) -> str:
66         """
67         Extracts selected segments from a raw video and stitches them together
68         into a seamless highlights file, applying end_padding to prevent abrupt endings.
69         """
70         if not segments:
71             raise ValueError("No highlight segments provided to compile.")
72
73         if not os.path.exists(raw_video_path):
74             raise FileNotFoundError(f"[COMPILER ERROR] Source video file not found:
75 {raw_video_path}")
76
77         output_path = os.path.join(self.output_dir, output_filename)
78
79         # Probe video duration so we can detect out-of-range segments
80         video_duration = self._get_video_duration(raw_video_path)
81         if video_duration > 0:
82             logger.info(f"Source video duration: {video_duration:.2f}s")
83         else:
84             logger.warning("Video duration unknown. Cannot validate segment boundaries
85 against video length.")
86
87         # We will create temporary clips and stitch them using FFmpeg concat demuxer
88         temp_clips = []
89         skipped_segments = []
90
91         # Create a temp directory within output_dir
92         with tempfile.TemporaryDirectory(dir=self.output_dir) as tmp_dir:
93             logger.info(f"Trimming {len(segments)} segments
94 (begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")
95             for idx, (raw_start, raw_end) in enumerate(segments):
96                 # Normalize timestamps to float seconds ? handles float, int,
97                 # plain numeric strings, and colon-separated formats like "MM:SS"
98                 start = self._parse_time(raw_start)
99                 end = self._parse_time(raw_end)
100                 segment_label = f"Segment #{idx+1}/{len(segments)} [{start:.2f}s -
101 {end:.2f}s]"
102
103                 # Validate the segment times
104                 if start < 0:
105                     logger.warning({segment_label}: start_time is negative
106 ({start:.2f}s). Clamping to 0.)
107                     start = 0.0
108
109                 if start >= end:
110                     logger.error({segment_label}: start_time ({start:.2f}s) >= end_time
111 ({end:.2f}s). Skipping invalid segment.)

```

39. assistant (2026-06-10T05:37:46.408Z)

Lines 100/104 of compiler.py look syntactically suspicious ? let me verify the file even compiles.

40. user (2026-06-10T05:37:48.994Z)

```

File "backend/pipeline/stitcher/compiler.py", line 100
    logger.warning({segment_label}: start_time is negative ({start:.2f}s). Clamping to 0.)

```

SyntaxError: invalid decimal literal

MAIN-OK

```

    segment_label = f"Segment #{idx+1}/{len(segments)} [{start:.2f}s - {end:.2f}s]"

```

```

        # Validate the segment times
        if start < 0:
            logger.warning({segment_label}: start_time is negative ({start:.2f}s).
Clamping to 0.)
            start = 0.0

        if start >= end:
            logger.error({segment_label}: start_time ({start:.2f}s) >= end_time
({end:.2f}s). Skipping invalid segment.)
            skipped_segments.append((idx, start, end, "start >= end"))
            continue

        # Check if segment extends beyond video duration
        if video_duration > 0:
            if start >= video_duration:

```

41. assistant (2026-06-10T05:37:57.265Z)

Critical finding confirmed ? compiler.py has a SyntaxError. Now checking whether main.py imports it at module level (meaning the app can't even start).

42. user (2026-06-10T05:38:04.490Z)

```

1:import os
2:import sys
3:import argparse
4:import json
5:import logging
6:import uvicorn
7:from dotenv import load_dotenv
18:from fastapi import FastAPI, HTTPException
19:from fastapi.middleware.cors import CORSMiddleware
20:from pydantic import BaseModel
21:from typing import List, Dict, Any, Optional
23:from fastapi.staticfiles import StaticFiles
24:from fastapi.responses import HTMLResponse
25:from backend.pipeline.validators.base import registry
26:from backend.infrastructure.database import Database
27:from backend.pipeline.segmenters.base import segmenter_registry, BasePlaySegmenter
28:from backend.pipeline.stitcher.compiler import VideoCompiler
29:from backend.utils.hashing import compute_video_id # canonical file-identity hashing
30:from backend.config import load_config as load_app_config, AppConfig # new typed config
31:from backend.results import Result, Failure # standardized error/result contract
32:from backend.reporting import emit_indexer_report
62:# New code should import load_app_config / AppConfig from backend.config directly.
229:    import cv2
300:        import subprocess
            from backend.pipeline.stitcher.compiler import VideoCompiler
File "C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\stitcher\compiler.py", line 100
    logger.warning({segment_label}: start_time is negative ({start:.2f}s). Clamping to 0.)
                                                                    ^

```

SyntaxError: invalid decimal literal

43. user (2026-06-10T05:38:24.656Z)

```

7fbbbeb0 feat(logging): standardize to logging in hot paths + update docs
8615534 refactor: adopt approved layered directory structure
8615534 2026-06-07 refactor: adopt approved layered directory structure
7fbbbeb08 (avidullu 2026-06-07 100)          logger.warning({segment_label}:
start_time is negative ({start:.2f}s). Clamping to 0.)

```

44. user (2026-06-10T05:38:43.720Z)

Tests package

```
=== loc ===
3567
=== importorskip/env ===
tests/test_reporting.py:70:# (see test_reporting_harvest.py which uses pytest.importorskip).
Those tests
tests/test_reporting_harvest.py:10:pytest.importorskip("obsreport")
tests/test_reporting_spec_faq.py:13:pytest.importorskip("obsreport")
=== validators cv2 ===
1:import cv2
2:from backend.pipeline.validators.base import VideoValidator, ValidationResult
3:from typing import Dict, Any
=== compile-all backend ===
  File "backend\pipeline\stitcher\compiler.py", line 100
    logger.warning({segment_label}: start_time is negative ({start:.2f}s). Clamping to 0.)
                                                                    ^
SyntaxError: invalid decimal literal

compileall-exit=0
```

45. assistant (2026-06-10T05:39:43.054Z)

All audit steps are complete. Compiling the structured report.

46. user (2026-06-10T05:41:01.020Z)

Structured output provided successfully

47. assistant (2026-06-10T05:41:08.879Z)

Audit complete ? structured output delivered. Key result: the suite is green (254 passed, 19.9s) but `backend/pipeline/stitcher/compiler.py:100` has a shipped SyntaxError that makes the whole FastAPI app unimportable, undetected because no test imports `backend.main`, the compiler, `wasb\_hybrid`, `motion`, or any concrete validator ? and no CI exists to run anything.